MINOR PROJECT 4 REPORT

PREDICTIVE CLASSIFICATION USING RANDOM FOREST ENSEMBLES

Project Title : Credit Card Fraud Detection

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ABSTRACT

This project focuses on predictive classification using the Random Forest algorithm for credit card fraud detection. The main objective is to classify transactions as genuine or fraudulent using machine learning techniques.

The project uses the Credit Card Fraud Detection dataset from Kaggle. The workflow includes data cleaning, exploratory data analysis, handling class imbalance using SMOTE, training a Random Forest classifier, and evaluating its performance using precision, recall, F1-score, confusion matrix, and ROC-AUC.

In the current experiment, the model achieved a ROC-AUC score of 0.9624, fraud precision of 0.89, and fraud recall of 0.78. These results show that the model can identify many fraudulent transactions, although some fraud cases remain undetected.

1. INTRODUCTION

Credit card fraud is a major problem in digital financial transactions. Fraud detection systems use machine learning techniques to identify suspicious transactions and distinguish them from genuine transactions.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to make predictions. It is useful for classification tasks because it can learn relationships between input features and the target class.

This project applies the Random Forest algorithm to credit card transaction data. The model is trained on historical transaction records and evaluated using multiple classification metrics to understand its strengths and limitations.

2. PROJECT OBJECTIVES

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The main objective of this project is to develop a machine learning classification model using the Random Forest algorithm to detect fraudulent credit card transactions.

The specific objectives of the project are:

1. To select and analyze a suitable dataset from Kaggle.
2. To inspect and clean the dataset by identifying missing values and duplicate records.
3. To perform Exploratory Data Analysis (EDA) to understand the distribution of genuine and fraudulent transactions.
4. To analyze relationships between selected features using a correlation heatmap.
5. To divide the dataset into training and testing sets using an 80:20 ratio.
6. To handle class imbalance using the SMOTE technique.
7. To train a Random Forest classification model.
8. To evaluate the model using a confusion matrix, precision, recall, F1-score, and ROC-AUC.
9. To identify important features contributing to model predictions.

3. DATASET DESCRIPTION

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The dataset used in this project is the Credit Card Fraud Detection dataset obtained from Kaggle. It contains transaction records that can be used to

train a machine learning model for identifying potentially fraudulent transactions.

The original dataset contains 284,807 rows and 31 columns. It includes 30 input features and one target column named Class.

The target variable represents two transaction categories:

- **Class 0:** Genuine transactions.
- **Class 1:** Fraudulent transactions.

Dataset Summary

Missing Values	Value
Dataset Source	Kaggle
Original Records	284,807
Total Columns	31
Input Features	30
Target Variable	Class
Missing Values	0
Duplicate Rows Identified	1,081

After removing duplicate rows, the dataset contains 283,726 records. The cleaned dataset contains 283,253 genuine transactions and 473 fraudulent transactions. This indicates a significant class imbalance, which must be considered during model training and evaluation.

4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is performed to understand the dataset, identify class imbalance, and examine relationships between the input features.

The dataset contains genuine and fraudulent credit card transactions. After removing duplicate records, the cleaned dataset contains 283,253 genuine transactions and 473 fraudulent transactions.

The number of genuine transactions is much higher than the number of fraudulent transactions. Therefore, the dataset is highly imbalanced. This imbalance can affect model training and makes it important to evaluate fraud-specific precision, recall, and F1-score rather than relying only on accuracy.

A correlation heatmap is also generated to examine relationships among selected features and the target variable.

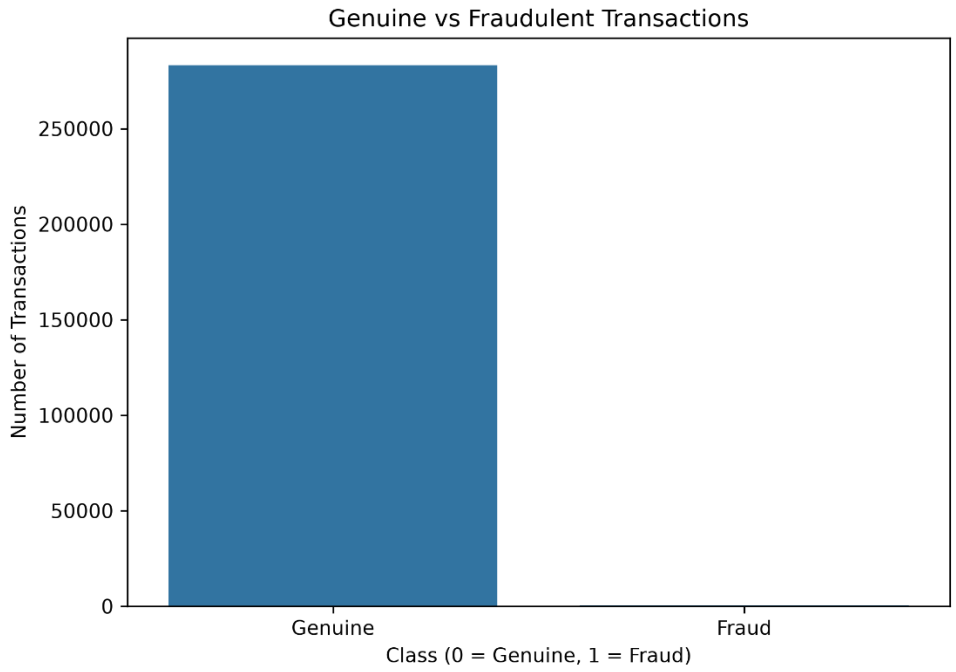


Figure 1: Distribution of Genuine and Fraudulent Transactions.

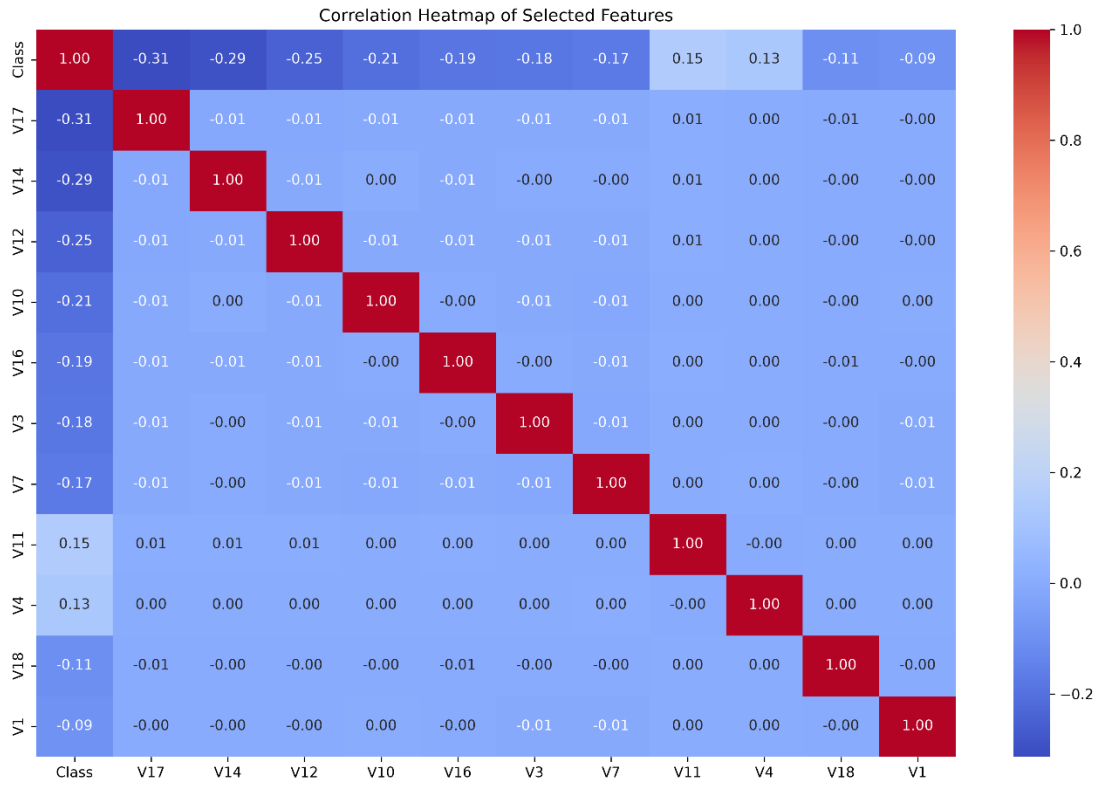


Figure 2: Correlation Heatmap of Selected Features.

5. DATA PREPROCESSING

Data preprocessing is an important step before training a machine learning model. It prepares the dataset for reliable model development and evaluation.

5.1 Handling Missing Values

The dataset was checked for missing values. The original dataset contained zero missing values, so no missing-value imputation was required for this dataset.

5.2 Duplicate Removal

The dataset initially contained 1,081 duplicate rows. These rows were removed from the working dataset, resulting in 283,726 records.

5.3 Feature and Target Selection

The Class column was selected as the target variable. The remaining columns were used as input features for classification.

5.4 Train-Test Split

The cleaned dataset was divided into training and testing sets using an 80:20 ratio. Stratified splitting was used to preserve the class distribution in both sets.

5.5 Handling Class Imbalance

The Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data to balance the genuine and fraudulent transaction classes.

The test dataset was kept separate from SMOTE resampling and used for evaluating the trained model.

5.6 Feature Scaling and Encoding

The input features are numerical, so categorical encoding was not required. StandardScaler was not used in the final Random Forest model because Random Forest does not generally require feature scaling.

6. RANDOM FOREST MODEL DEVELOPMENT

Random Forest is a supervised machine learning algorithm that uses multiple decision trees to perform classification. It combines the predictions of different trees to produce the final prediction.

In this project, Random Forest is used to classify credit card transactions into two categories: genuine and fraudulent.

6.1 Model Configuration

The Random Forest Classifier was implemented using the Scikit-learn library in Python.

The selected hyperparameters are:

Hyperparameter	Value	Description
n_estimators	100	Number of decision trees
max_depth	20	Maximum depth of each tree
random_state	42	Ensures reproducibility
n_jobs	-1	Uses available CPU cores

6.2 Justification of Hyperparameters

Number of Trees (n_estimators = 100):
The model uses 100 decision trees to combine multiple predictions and improve classification performance.

Maximum Depth (max_depth = 20):
The maximum depth limits the growth of each tree and controls model complexity.

Random State (random_state = 42):
This parameter helps produce reproducible results when the code is executed again with the same environment and data.

6.3 Model Training

The Random Forest model was trained using the training dataset after applying SMOTE. The testing dataset was kept separate to evaluate the model on unseen examples.

The trained model was saved as models/fraud_model.pkl for future use.

7. MODEL EVALUATION AND RESULTS

The performance of the Random Forest classification model was evaluated using a separate test dataset. The evaluation metrics include accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

The test dataset contained 56,746 transactions, including 56,651 genuine transactions and 95 fraudulent transactions. The results help assess the model's ability to identify fraudulent transactions while limiting false alarms.

7.1 Classification Report

metric	genuine	fraud	descrption
precision	**1.00**	**0.89**	**Correction of positive predictions**
recall	**1.00**	**0.78**	**Proportion of actual cases detected**
F1-score	**1.00**	**0.83**	**Harmonic mean of precision and recall**
support	**56,651**	**95**	**Number of actual test examples**

ROC-AUC Score: 0.9624

7.2 Confusion Matrix

actual / Predicted	Genuine	Fraud
Genuine	**56,642**	**9**
Fraud	**21**	**74**

The confusion matrix shows that the model correctly classified 56,642 genuine transactions and detected 74 fraudulent transactions. It incorrectly flagged 9 genuine transactions as fraud and failed to detect 21 fraudulent transactions.

The fraud recall of 0.78 indicates that the model detected approximately 78% of actual fraud cases in the test dataset. Further improvements may help reduce missed fraud cases.

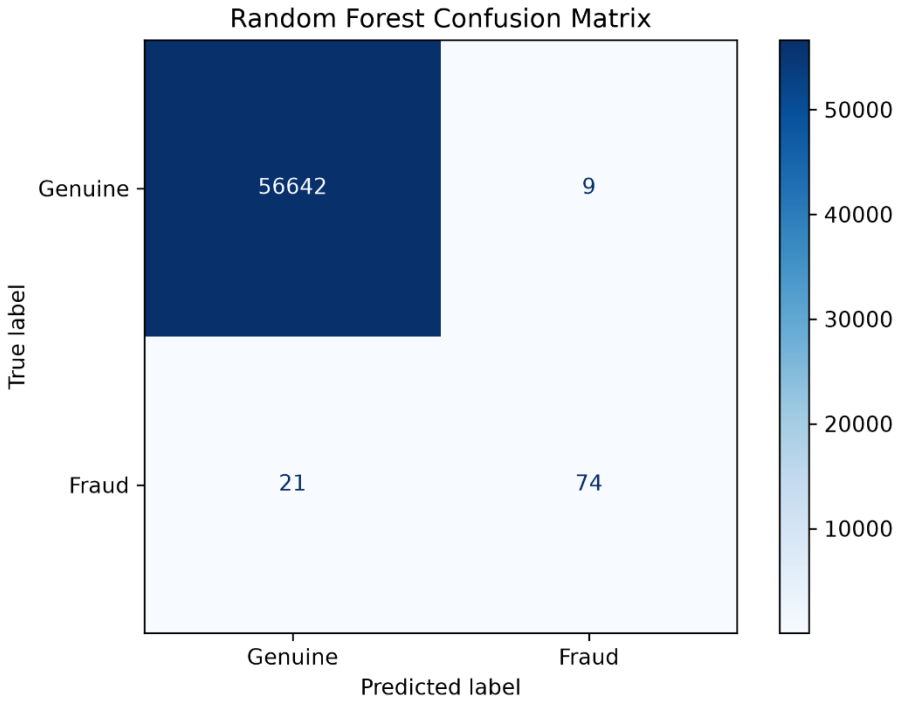


Figure 3: Confusion Matrix of the Random Forest Classifier.

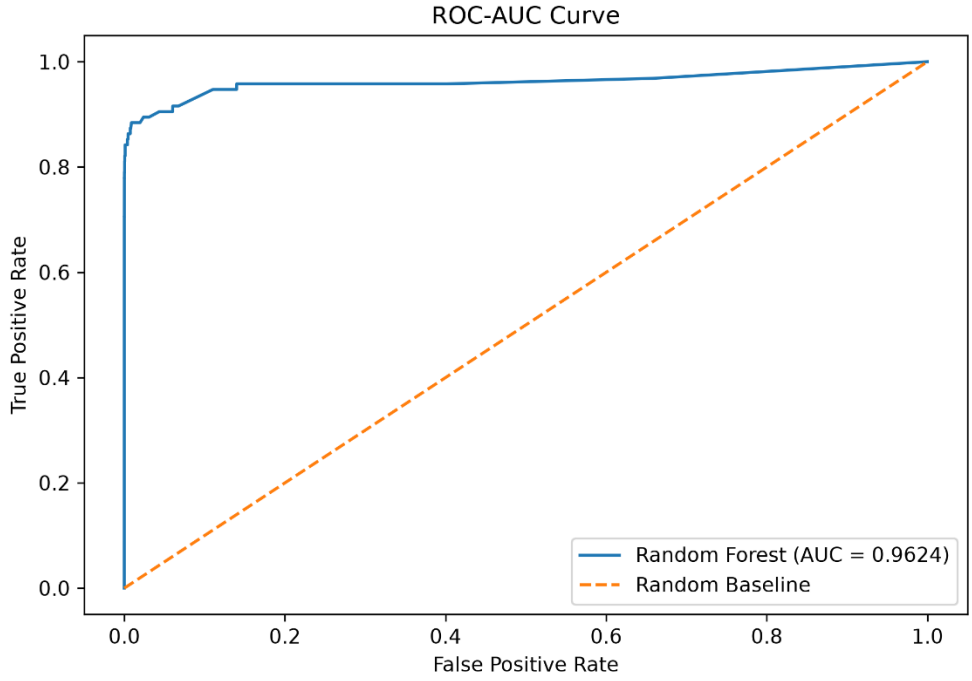


figure 4: ROC-AUC Curve of the Random Forest Classifier.

The ROC curve illustrates the relationship between the true-positive rate and false-positive rate at different classification thresholds. The model achieved a ROC-AUC score of 0.9624, indicating good discrimination between genuine and fraudulent transactions.

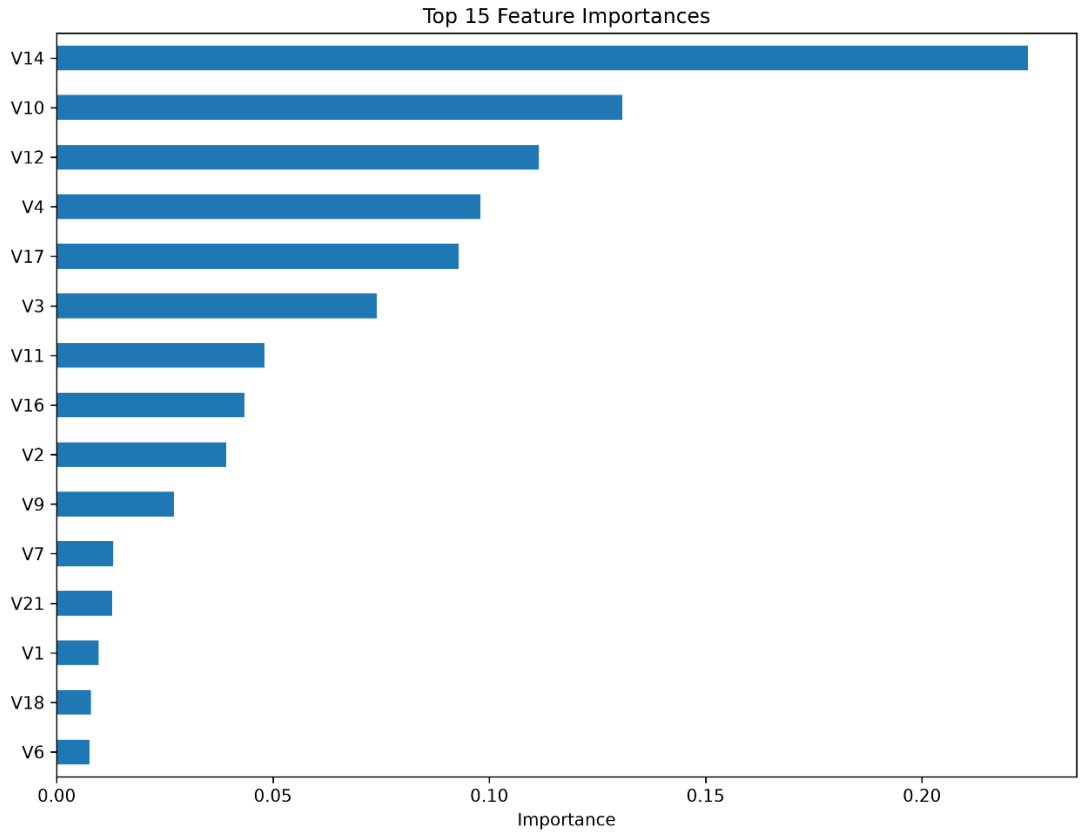


Figure 5: Feature Importance in the Random Forest Model.

Feature importance identifies the input variables that contributed most to the Random Forest model's predictions. These values help understand the model's decision process, but they do not establish causal relationships.

Accuracy: Approximately 99.95%.

8. LIMITATIONS

The Random Forest model performed well on the test dataset, but some limitations remain.

1. Missed Fraud Cases: The model failed to detect 21 fraudulent transactions out of 95 actual fraud cases in the test dataset.

2. Class Imbalance: The original dataset contains significantly more genuine transactions than fraudulent transactions.
3. False Positives: The model incorrectly classified 9 genuine transactions as fraudulent.
4. Hyperparameter Tuning: The model uses selected parameters, but extensive hyperparameter optimization has not been performed.
5. Generalization: The results are based on a particular dataset and test split. Performance on new transactions may differ.

10. CONCLUSION

This project successfully implemented a Random Forest classification model for credit card fraud detection using Python and machine learning libraries.

The model achieved a ROC-AUC score of 0.9624, fraud precision of 0.89, and fraud recall of 0.78. It correctly detected 74 fraudulent transactions in the test dataset but missed 21 actual fraud cases.

The results demonstrate that Random Forest can help identify potentially fraudulent transactions. However, further improvements are necessary to reduce missed fraud cases and improve the model's reliability.

FUTURE SCOPE

The project can be improved in the following ways:

1. Perform hyperparameter tuning to identify better model configurations.
2. Compare Random Forest with Gradient Boosting and other classification algorithms.
3. Adjust the classification threshold to study the precision-recall trade-off.
4. Evaluate the model using cross-validation.

5. Test the model on additional transaction data to assess its generalization ability.

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2. Scikit-learn Documentation, *RandomForestClassifier*.
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4. Imbalanced-learn Documentation, *SMOTE — Synthetic Minority Over-sampling Technique*.